PROPORTIONAL-INTEGRAL-RESONANT CONTROL FOR THE PERIODIC DISTURBANCE MINIMIZATION OF THE PMSM

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Abstract

This paper presents the periodic disturbance minimization technique by using the proportional-integral-resonant (PIR) control scheme for the permanent magnet synchronous motors (PMSM). Flux harmonics, cogging torque, and the current sampling errors are the prominent sources of the periodic disturbances in the PMSM. The conventional proportional-integral (PI) control scheme is widely employed in the cascaded control structure to regulate the PMSM for the specified speed and control references. However, the conventional PI controllers cannot effectively minimize the periodic disturbances which produce the speed and torque ripples. Therefore, the resonant controller is coupled with the PI controller in the current control loop to minimize the periodic disturbances of the PMSM. The bandwidth of the resonant controllers is determined by the selection of the suitable gains by using the bode plot analysis. In addition, the impact of the speed change on the bandwidth of the resonant controllers is also investigated in detail by using the bode plot analysis. Finally, the comparative results of the conventional PI controller are compared with the proposed PIR current control to highlight its control efficacy. The simulation results demonstrate the better performance of the PIR control for the wide operating range compared to the conventional PI controller.

1 Introduction

High power density, good dynamic performance, and low maintenance cost are the prominent advantages of the permanent magnet synchronous motor (PMSM) that encourages its industrial application in the robots, computer numerical control (CNC) machine tools, medical equipment, and electric vehicles [1]. However, the performance of the PMSM is deteriorated due to the presence of unwanted periodic disturbances such as cogging torque, current sampling error, and flux harmonics. These periodic disturbances can cause high torque vibrations at low/medium speed range and large acoustic noise at high-speed range. The torque ripples are filtered out at high speeds due to inertia and friction, the higher order torque producing harmonics can be disregarded. The torque ripple minimization techniques are generally categorized as the design-based techniques and control-based techniques [2]. First, the design-based torque ripple minimization techniques of the PMSM drives are developed by modifying the various shapes of the stator and rotor such as skewing magnets, skewing slots, addition of supplementary teeth/slots, and changing pole arc to pole pitch ratio. However, the design modifications increase the manufacturing cost, reliability issues, high maintenance cost, and complex control design. Second, the control-based torque ripple minimization techniques employ the advanced control modifications without modifying the design of the motor which makes it an optimum choice for the existing motors operating in the industry.

Most of the industrial applications of the PMSM drives employ the cascaded control structure that consists of the inner current loop and outer speed loop [3]. The cascaded control structure based on the field-oriented control (FOC) and direct torque control (DTC) are employed for the application of the PMSM in the above-mentioned applications. The classical proportional-integral (PI) control is widely employed for the implementation of the FOC-based cascaded control structure due to its relatively simple implementation and gain tuning procedures. However, the PI controller have the linear control structure which cannot accurately control the nonlinear time varying PMSM drives. In particular, the presence of the unwanted periodic disturbances in the PMSM drives significantly deteriorates the control performance of the PI controllers. As the PI control do not have any compensating term for the periodic disturbances, its performance is highly sensitive to the unwanted periodic disturbances which results in the high speed and torque ripples.

Numerous control modifications are made in the PI cascaded control structure to minimize the negative effects of the periodic disturbances. The PI controllers are coupled with the iterative learning control (ILC) [4] and repetitive control to mitigate the effects of the periodic disturbances.

ILC is an error-correction algorithm equipped with the memory which saves the output and error information of the previous time intervals. The ILC reduces the periodic disturbances by learning from the previous iterations data stored in its memory and generates a feedforward control input. The feedforward control input provided by the ILC is coupled with the PI controller to enhance the overall control performance of the PMSM [4]. However, the performance of ILC is improved by increasing the number of iterations which increases the overall computational burden. The stored data in the memory of the ILC requires additional storage space which increases the implementation cost of the ILC.

The repetitive control is also combined with the PI control to design the plug-in-repetitive control for the torque ripple minimization of the PMSM. The repetitive controller for current loop is designed to achieve the zero static error tracking of the periodic disturbances but the repetitive controller requires a long-time interval to reach steady-state condition. Moreover, the repetitive control requires careful gain selection that results in the slow dynamic response of the drive system. For ILC and repetitive control, the tracking error diverges after converging with an increase in the number of iterations [5].

To minimize the effects of the periodic disturbances, this paper presents the proportional-integral-resonant controller (PIR) for the PMSM drive. The resonant controller (RC) is coupled in parallel with the PI controller in the current loop to formulate the PIR control. The PMSM normally encounter the periodic disturbances of the 6th order and 12th order harmonics of the fundamental frequency. Therefore, the two PIR controllers are designed to minimize the 6th order and 12th order harmonics of the PMSM drives. The PIR controller minimize the effects of the periodic disturbances by achieving the zero static tracking error of the periodic components. The detailed analysis on the selection of the PIR gains is presented and the effects of the speed change on the bandwidth of the PIR controller is presented by using the bode plots.

2. Dynamic Model of PMSM

The dynamic model of the PMSM in the synchronously rotating *dq*-reference frame is as follows [6]:

$$\begin{aligned}
\frac{di_{qs}}{dt} &= \frac{1}{L_q}\left(v_{qs} - R_s i_{qs} - \lambda_{dm}\omega_r - L_d\omega_r i_{ds}\right)\\
\frac{di_{ds}}{dt} &= \frac{1}{L_d}\left(v_{ds} - R_s i_{ds} + L_q\omega_r i_{qs} + \lambda_{qm}\omega_r\right)\\
\frac{d\omega_r}{dt} &= \frac{p}{2J}\left(\tau_e - \frac{2B}{p}\omega_r - \tau_L\right)\\
\tau_e &= 1.5\frac{p}{2}\left(\lambda_m i_{qs} + (L_d - L_q)i_{ds}i_{qs}\right)
\end{aligned} \tag{1}$$

where τ_e is the electromagnetic torque, τ_L is the load torque considered as an unknown disturbance that changes slowly during a small sampling period (T_s), p is the number of poles, R_s is the stator resistance, L_d and L_q are the *d*-axis and *q*-axis inductances, J is the rotor inertia, B is the viscous friction coefficient, λ_{dm} and λ_{qm} are the *d*-axis and *q*-axis magnet flux linkage, ω_r is the electrical rotor speed, i_{ds} and i_{qs} are the *d*-axis and *q*-axis stator currents, and v_{ds} and v_{qs} are the *d*-axis and *q*-axis voltages, respectively. The sources of the periodic disturbances in the PMSM are presented as follows:

2.1 Flux Harmonics

The ideal PMSM has a pure sinusoidal flux linkage which implies that the *q*-axis flux λ_{qm} is zero and λ_{dm} has the *dc* component λ_m in the synchronously rotating *dq* reference frame. For the actual PMSM, it is difficult to obtain ideal sinusoidal flux distribution due to magnetic saturation and manufacturing restriction, which produce the flux harmonics. In the PMSM, the λ_{qm} and λ_{dm} have the 6th order harmonics which are presented as follows [7]:

$$\begin{aligned}
\lambda_{md} &= \lambda_m + \sum_{n=1}^{\infty}\lambda_{dhn}\cos(6n\theta_e)\\
\lambda_{mq} &= \sum_{n=1}^{\infty}\lambda_{qhn}\sin(6n\theta_e)
\end{aligned} \tag{2}$$

where n, λ_m, λ_{dhn}, and λ_{qhn} represents the positive integer, dc component of the *d*-axis flux, amplitude of the 6*n*th order flux harmonics of the *d*- and *q*-axes, respectively. The back EMF contains the 6*n*th order harmonics which acts as a disturbance in the current loop. The flux harmonics affect the generation of the electromagnetic torque τ_e and produce the 6*n*th periodic ripples in the torque and speed.

2.2 Current Scaling Error Harmonics

The current measurement errors are among the major sources of the periodic disturbances in the PMSM. The actual *q*-axis current i_{qs} is represented as [6]:

$$\begin{aligned}
i_{qs} &= i_{qs0} + i_{qs,offset}\\
&= i_{qs0} + \frac{2}{3}\cos\left(\theta_e + \varphi\right)\sqrt{\Delta i_a^2 + \Delta i_a \Delta i_b + \Delta i_b}
\end{aligned} \tag{3}$$

where i_{qs0} represents the dc component of the actual i_{qs}, Δi_a and Δi_b represents the dc offsets in the current sensors of the stator currents i_a and i_b, $\theta_e = 2\pi f_s t$, f_s is the fundamental frequency, t is the time, φ represents the constant displacement depending on the Δi_a and Δi_b, $i_{qs,offset}$ represent the *q*-axis current harmonics.

2.3 Deadtime Effect Harmonics

To avoid the short circuit in the switches of the voltage source inverter, the fixed deadtime T_d is inserted in the sampling period of the pulse-width modulation (PWM). However, the presence of the deadtime results in the unwanted voltage harmonics that can represented as [6]:

$$\begin{aligned}
v_{hd} &= \frac{4T_d V_{dc}}{\pi T_s}\sum_{n=1}^{\infty}\left[\frac{12n}{36n^2-1}\sin\left(6n\theta_e\right)\right]\\
v_{hq} &= \frac{4T_d V_{dc}}{\pi T_s}\left\{-1 + \sum_{n=1}^{\infty}\left[\frac{2}{36n^2-1}\cos\left(6n\theta_e\right)\right]\right\}
\end{aligned} \tag{4}$$

where v_{hd} and v_{hq} are the *dq*-axis voltage harmonics, V_{dc} is the dc-link voltage, T_s is the sampling period.

2.4 Cogging Torque Harmonics

The cogging torque is the torque produced when the permanent magnet is approaching or leaving the stator teeth, and it is independent of the stator current. The commonly used Fourier expression of the cogging torque is as follows:

$$\tau_{cog} = \sum_{n=1}^{\infty} \tau_n \sin(nN_{co}\theta_e) \tag{5}$$

where N_{co} is the least common multiple between motor slots and poles, and τ_n is the amplitude of the nth component of the Fourier series. The cogging torque produces the periodic disturbances in the speed loop.

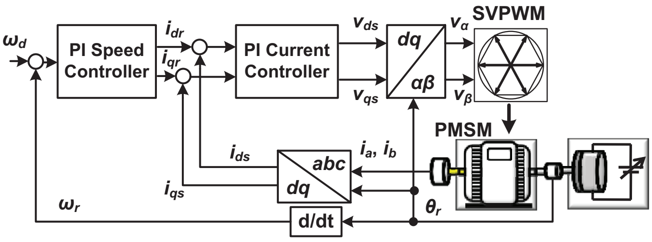


Fig. 1 Conventional PI-PI cascaded control configuration of a PMSM drive

3. Periodic Disturbance Minimization by PIR

This section presents the conventional PI controller for the PMSM drives. Next the PIR controller for the current loop are introduced to reduce the periodic disturbances of the PMSM drives.

3.1 Cascaded PI-PI Controller

In the most industrial applications, the cascaded PI control structure is employed due to its simple control configuration, easy design, and satisfactory performance. Fig. 1 presents the conventional cascaded PI-PI control configuration which has the outer speed loop and the inner current loop. The PI speed controller which generates the q-axis reference current is presented as follows:

$$i_{qr} = K_{ps}(\omega_d - \omega_r) + \frac{K_{is}}{s}(\omega_d - \omega_r) \tag{6}$$

where $K_{ps} = \omega_{sc}J/1.5p^2\lambda_{dm}$ and $K_{is} = \omega_{sc}^2 J/1.5p^2\lambda_{dm}$ are the control gains for the PI speed controller, ω_d is the reference for the speed PI controller, i_{qr} is the reference for the PI current controller, s is the Laplace operator, $\omega_{sc}= 2\pi f_{sc}$, and f_{sc} is the bandwidth of the PI speed controller.

The maximum torque per ampere (MTPA) control is used for the torque generation in the constant torque region. The reference d-axis current i_{dr} is calculated as:

$$i_{dr} = \frac{L_d - L_q}{\lambda_{md}} i_{qr}^2 \tag{7}$$

In the inner loop of the PI-PI cascade structure, the dq-axis PI current controllers are employed to design the control voltage commands in the rotating dq-axis reference frame as:

$$\begin{aligned} v_{ds} &= K_{pcd}(i_{dr} - i_{ds}) + \frac{K_{icd}}{s}(i_{dr} - i_{ds}) \\ v_{qs} &= K_{pcq}(i_{qr} - i_{qs}) + \frac{K_{icq}}{s}(i_{qr} - i_{qs}) \end{aligned} \tag{8}$$

where $K_{pcd} = \omega_{cc} L_d$, $K_{pcq} = \omega_{cc} L_q$, and $K_{icd} = K_{icq} = \omega_{cc} R_s$ are the control gains of the PI dq-axis current controllers, and $\omega_{cc}= 2\pi f_{cc}$ is the bandwidth of the PI current controller. Despite the simpler structure, the PI controllers are not effective for the periodic disturbances of the PMSM.

3.2 PIR Controller

The PIR controller is designed by connecting the resonant controllers in parallel with the PI controller. The resonant controller minimizes the effects of the periodic disturbances in the PMSM, and the PI controllers determines the dynamic performance of the system and produces the dc control output [5]. The resonant controller is presented as:

$$G(s) = \frac{k_r s}{s^2 + \omega_0^2} \tag{9}$$

where k_r is the resonant coefficient, and ω_0 is the resonant frequency. The PIR controller aims to make i_{qs} track i_{qr} accurately and reject the disturbances as well. The dead time voltage and back EMF are the main disturbances in the current loop which contain the 6nth order harmonic frequencies of the ac fundamental frequency components at the steady state. Therefore, the resonant terms with the resonant frequencies of $6n\omega_r$ are added in parallel with the PI terms to improve the current controller performance. The n = 2 is selected to maintain the balance between the complexity and performance, the higher order harmonics have little effect on the torque and speed ripples. The PIR controller for the i_{ds} has the similar structure.

The open loop gain including the PIR controller and PMSM is as follows:

$$G_{o\_cc} = \left[\left(K_{pcq} + \frac{K_{icq}}{s}\right) + \frac{k_r s}{s^2 + \omega_0^2}\right]\frac{1}{L_q\, s + R_s} \tag{10}$$

The closed loop transfer function for the PIR current controller is:

$$H_{o\_cc}(s) = \frac{G_{o\_cc}}{1 + G_{o\_cc}} \tag{11}$$

The PMSM is modelled as an electrical load which consists of the resistor and inductor in series. The bandwidth of the resonant controller is selected by using the suitable value of the k_r. Fig. 2 presents the PIR controller configuration with the 6th order and 12th order harmonics cancellation.

$$H_{o\_cc}(\mathrm{s}) = \frac{L_q\omega_{cc}s^3 + (R_s\omega_{cc} + k_r)s^2 + \omega_{cc}(n\omega_r)^2 L_q s + \omega_{cc}(n\omega_r)^2 R_s}{L_q s^4 + (R_s + L_q\omega_{cc})s^3 + \left(R_s\omega_{cc} + k_r + (n\omega_r)^2 L_q\right)s^2 + \left((n\omega_r)^2 R_s + \omega_{cc}(n\omega_r)^2 L_q\right)s + \omega_{cc}(n\omega_r)^2 R_s} \tag{12}$$

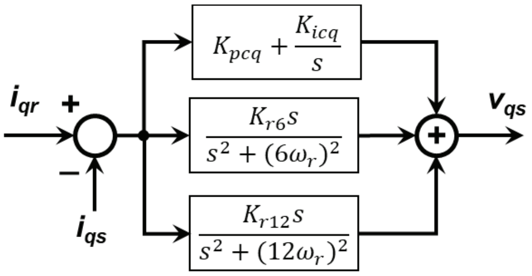
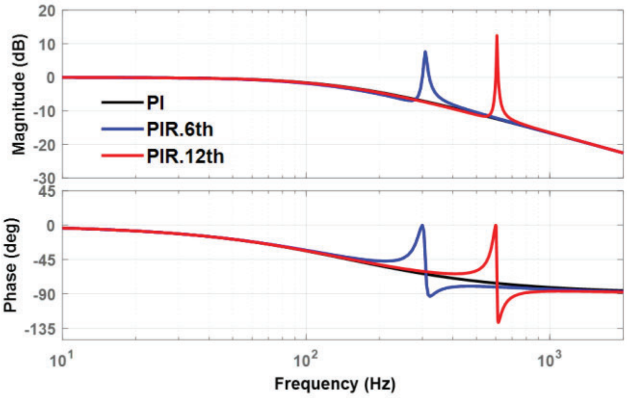


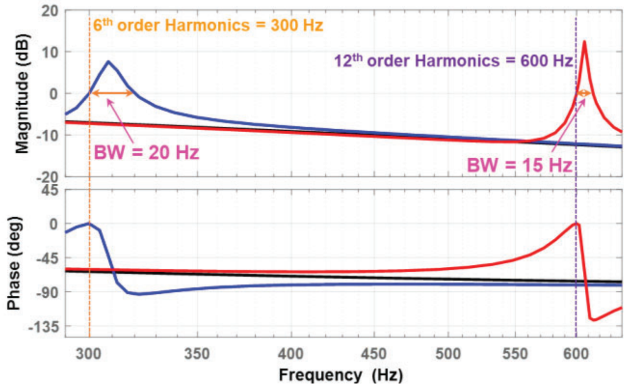
Fig. 2 PIR control configuration for the current loop of the PMSM drive

4. Simulation Verifications

The MATLAB/Simulink based simulation model is designed to verify the performance of the PI control scheme and PIR control scheme. The prototype IPMSM drive has the following nominal parameters: q-axis inductance L_q = 113.9 mH, d-axis inductance L_d = 75.0 mH, stator resistance R_s = 2.5 Ω, magnetic flux linkage λ_m = 0.193 V·s/rad, equivalent rotor inertia J = 0.00042 kg·m², viscous friction coefficient B = 0.0001 N·m·s/rad, number of poles p = 4, dc-link voltage V_{dc} = 295 VDC, sampling time T_S = 200 µs, dead-time T_{dead} = 2 µs, rated power P_{rated} = 390 W, rated torque $T_{L\text{-}rated}$ = 1.5 N·m, rated speed $\omega_{r\text{-}rated}$ = 2500 r/min, and space vector PWM (SVPWM) frequency f_s = 5 kHz. The current bandwidth (i.e., f_{cc} = 150 Hz) in the inner control loop is set to 10 times higher value than the speed bandwidth (i.e., f_{sc} = 15 Hz) in the outer control loop based on pole placement design.

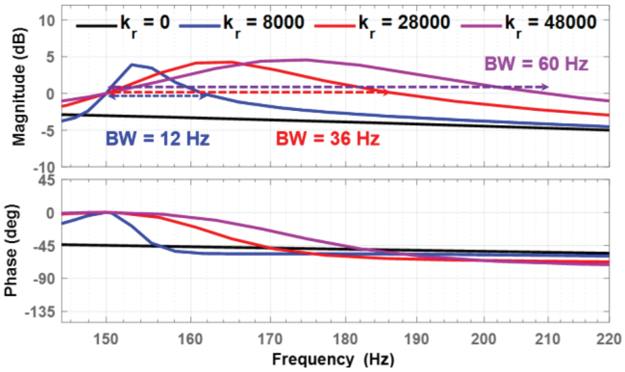


(a)

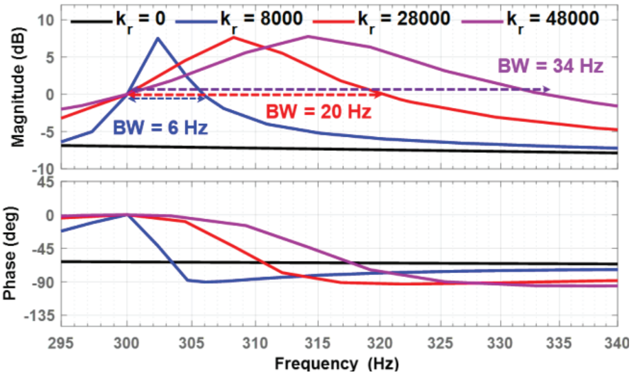


(b)

Fig. 3 Bode plot of the closed loop PIR control system at the 50 Hz fundamental frequency (a) Bode plot for the whole frequency range (b) Zoomed-in bode plot near 6th and 12th order harmonic frequency



(a)



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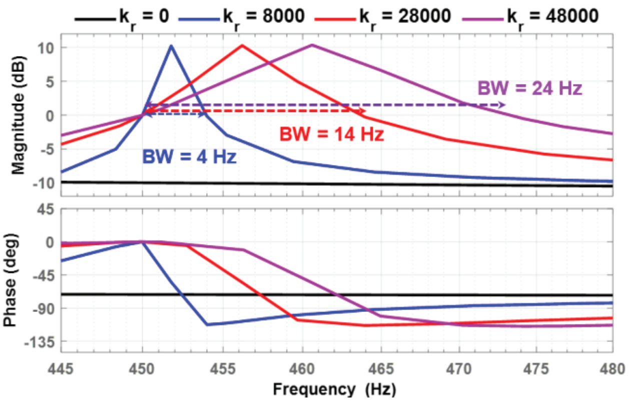
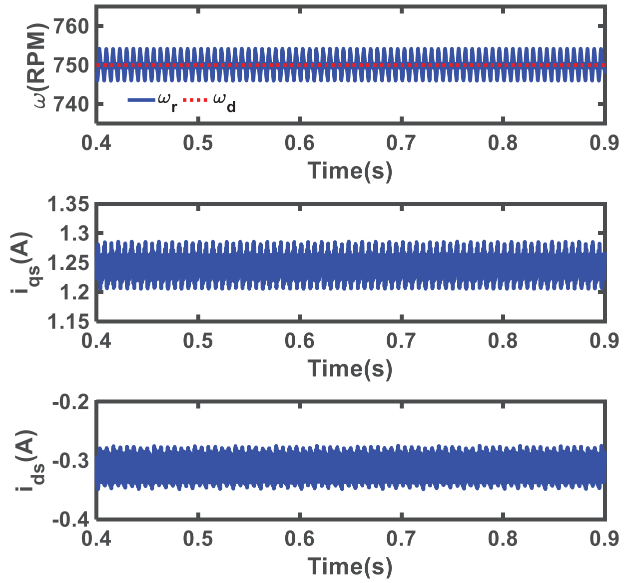


(c)

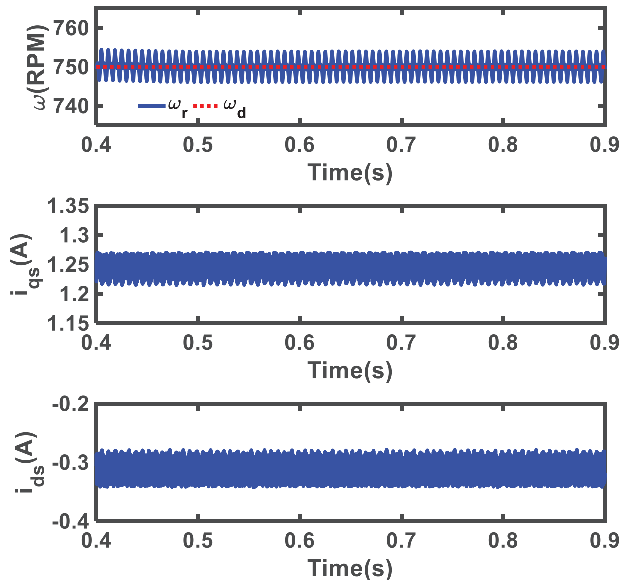
Fig. 4 Bode plot of the closed loop PIR control system for 6th order harmonic at different values of k_r (a) 25 Hz fundamental frequency at 750 r/min (b) 50 Hz fundamental frequency at 1500 r/min (c) 75 Hz fundamental frequency at 2250 r/min

Fig. 3 presents the bode Bode plot magnitude and phase of the closed loop PIR control system at the 50 Hz fundamental frequency with k_{r6} = 28000 and k_{r12} = 39000 by using (12). Fig. 3(b) shows that the resonant controller with k_{r6} and k_{r12} have the 20 Hz and 15 Hz bandwidth (BW), respectively. The closed loop phase shift is 0 at the 300 Hz (6th order harmonic frequency) and 600 Hz (12th order harmonic frequency).

Fig. 4 presents the closed loop bode plot analysis for the 6th order PIR controller system at k_r = 0, 8000, 28000, 48000 operating under fundamental frequency of 25 Hz, 50 Hz, and 75 Hz. The k_r = 0 represents the PI controller without the resonant controller. The bandwidth of the resonant controller depends on the value of k_r. By increasing the value of the k_r, the width between the two zero crossing increases. This implies that the PMSM harmonics will be affected within the frequency range between two zero-crossing points. The high values of k_r makes the resonant controller more sensitive to the noise and poor transient response. For k_r = 28000, the bandwidth of the resonant controller is 36 Hz, 20 Hz and 14 Hz at the fundamental frequency of 25 Hz (750 r/min), 50 Hz (1500 r/min), and 75 Hz (2250 r/min). Fig. 4 highlights that the operating fundamental frequency and k_r are the key factors in determining the bandwidth of the resonant controllers.



(a)



(b)

Fig. 5 Comparative performance evaluation at 750 r/min with 0.75 Nm load (a) Conventional PI (b) Proposed PIR

Fig. 5 presents the comparative performance of the conventional PI and proposed PIR controller at 750 r/min with 0.75 Nm load. By using k_{r6} = 28000 and k_{r12} = 39000, the proposed PIR controller effectively minimize the harmonics in the i_{ds} and i_{qs} compared to the conventional PI controller which results in the smoother speed response.

5 Conclusion

This paper presents the PIR controller for the periodic disturbance minimization of the PMSM. The performance of the conventional cascaded PI-PI control configuration is degraded due to the presence of the flux harmonics, cogging torque, and the current sampling errors in the PMSM. To improve the performance of the cascaded PI-PI control configuration, two resonant controllers are added in the PIR control configuration to minimize the periodic ripples of the 6th order and 12th order harmonics. The bandwidth of the PIR controller is affected by the variable operating frequency and k_r gain. The comparative results verify the better performance of the PIR controller compared to the PI controllers for the harmonics rejection of the PMSM.

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